

Global Aviation Disruption Assessment 2026

Executive Case Study Report

Completed aviation-disruption assessment, business findings, and planning recommendations for the requesting team.

Scenario note: This is a simulated ICAO client assignment, not an official ICAO commission or endorsement.

Executive summary

The Middle East is the main concentration in this assessment. It accounts for 56.12% of airport-affected flights and 77.85% of modeled airline summary loss. This makes the region the first focus for contingency planning and financial scenario review.

The detailed records contain 2,200 cancellations affecting 575,710 passengers and 1,500 reroutes. Those reroutes add 2.20 million kilometres, USD 11.66 million in fuel cost, and 4,706.30 delay hours. Europe to Middle East is the leading reroute corridor, with 196 records.

The recommended priorities are to review the most exposed airports and directional corridors, prepare passenger-support plans for cancellation concentrations, and assess airline financial exposure using the separate summary and daily-estimate scenarios. The event-date comparison does not show a consistent increase across the operational measures, so it does not support a causal conclusion.

1 Business problem and objectives

The requesting team needs to understand where aviation disruption is concentrated and how it affects passengers, routes, airport activity, airspace access, and financial exposure. The assessment answers the 24 questions in the Business Requirements and turns the results into planning priorities.

Operational records cover late February to 7 June 2026. Financial amounts are undated modeled values. Airport and airspace counts describe separate reported exposures. These distinctions apply to the findings below and prevent misleading combined totals.

2 Operational and passenger findings

2.1 Cancellations affect a substantial passenger population

The 2,200 cancellation records contain 575,710 affected passengers. Monthly cancellation counts are 684 in March, 692 in April, and 650 in May. February and June are partial months and should not be compared with full months as if coverage were equal.

The most frequent supplied reason is security risk related to a conflict zone, with 222 records. Air traffic control congestion follows with 215, and temporary corridor suspension with 213. These categories help structure passenger-support and disruption-review discussions; they do not establish independently verified causes.

2.2 Reroutes add distance fuel cost and delay

Detailed reroute measure	Result
Reroute records	1,500
Additional distance	2,201,390 km
Additional fuel cost	USD 11,658,572.59
Delay burden	4,706.30 hours

The additional distance, cost, and delay give planning teams a clearer view of reroute burden than counts alone. The passenger total above belongs to cancellation records, not rerouted flights.

3 Geographic and corridor findings

3.1 Airport exposure is concentrated in the Middle East

The Middle East accounts for 20,116 of 35,844 airport-affected flights, or 56.12%. Asia Pacific follows with 4,354 and Europe with 3,323. Imam Khomeini International Airport has the largest airport total at 1,651, followed by Basra at 1,547 and Queen Alia at 1,510.

Airport duration and affected-flight rankings answer different questions. São Paulo-Guarulhos has the longest individual reported disruption at 71.7 hours, even though it does not lead affected-flight exposure. Planning reviews should consider both scale and duration.

3.2 Directional corridors have different burdens

Europe to Middle East has 196 reroutes, or 13.07% of all reroute records. It also has the largest regional-corridor burden: 294,387 additional kilometres, USD 1,566,712.76 extra fuel cost, and 603.90 delay hours. The reverse direction has 168 reroutes and should be reviewed separately.

Middle East to Europe has the most regional-corridor cancellations, with 251. Europe to Middle East follows with 250 and contains the largest cancellation-passenger sum, 67,417. These concentrations support corridor-specific planning rather than one undirected route total.

3.3 Airspace exposure requires a separate planning view

The closure records report 7,014 hours and 35,502 affected flights. Iraq has the largest closure duration at 1,140 hours, followed by Lebanon at 744 and Iran at 702. These figures are separate from airport exposure; adding the two flight totals would count an unknown amount of overlap.

4 Financial scenario findings

4.1 Summary loss is concentrated in Middle East airlines

The 70-airline summary reports USD 21.41 billion in modeled loss. Middle East airlines account for USD 16.67 billion, or 77.85%. Gulf Air has the largest modeled amount at USD 1.76 billion, followed by Oman Air at USD 1.63 billion and Kuwait Airways at USD 1.55 billion.

Flag carriers account for USD 13.81 billion, low-cost airlines for USD 5.19 billion, cargo airlines for USD 1.71 billion, and the private category for USD 0.71 billion. The groups contain different numbers of airlines, so total loss alone is not a measure of comparative airline performance.

4.2 Daily estimates describe a separate financial scenario

The 35-airline estimate reports USD 48.84 million in estimated daily loss and USD 27.61 million in estimated additional fuel cost. Emirates has the largest daily estimate at USD 4.20 million, followed by Qatar Airways at USD 3.10 million and El Al at USD 2.80 million.

Only 32 airlines are shared between the financial populations. The daily estimate is not a daily breakdown of the USD 21.41 billion summary. Financial planning should use each scenario separately, with its own airlines and assumptions.

The report also shows an 18.81% unweighted mean revenue-loss percentage and an estimated fuel cost of USD 36.61 thousand per estimated reroute. The estimated passenger-to-reroute measure is an exposure ratio, not aircraft occupancy.

5 Event context

The event data contains 82 records, including 18 military records. Within the common operational period from 28 February to 7 June, 12 dates contain military records and 88 do not. Mean cancellations are 21.08 on military-event dates and 22.07 on other dates. Mean reroutes are 15.08 and 14.99 respectively.

The measures do not move consistently in the same direction. This assessment therefore does not justify attributing the disruption pattern to military events alone. Event timelines provide context; operational and financial priorities are supported by the reported concentrations and burdens.

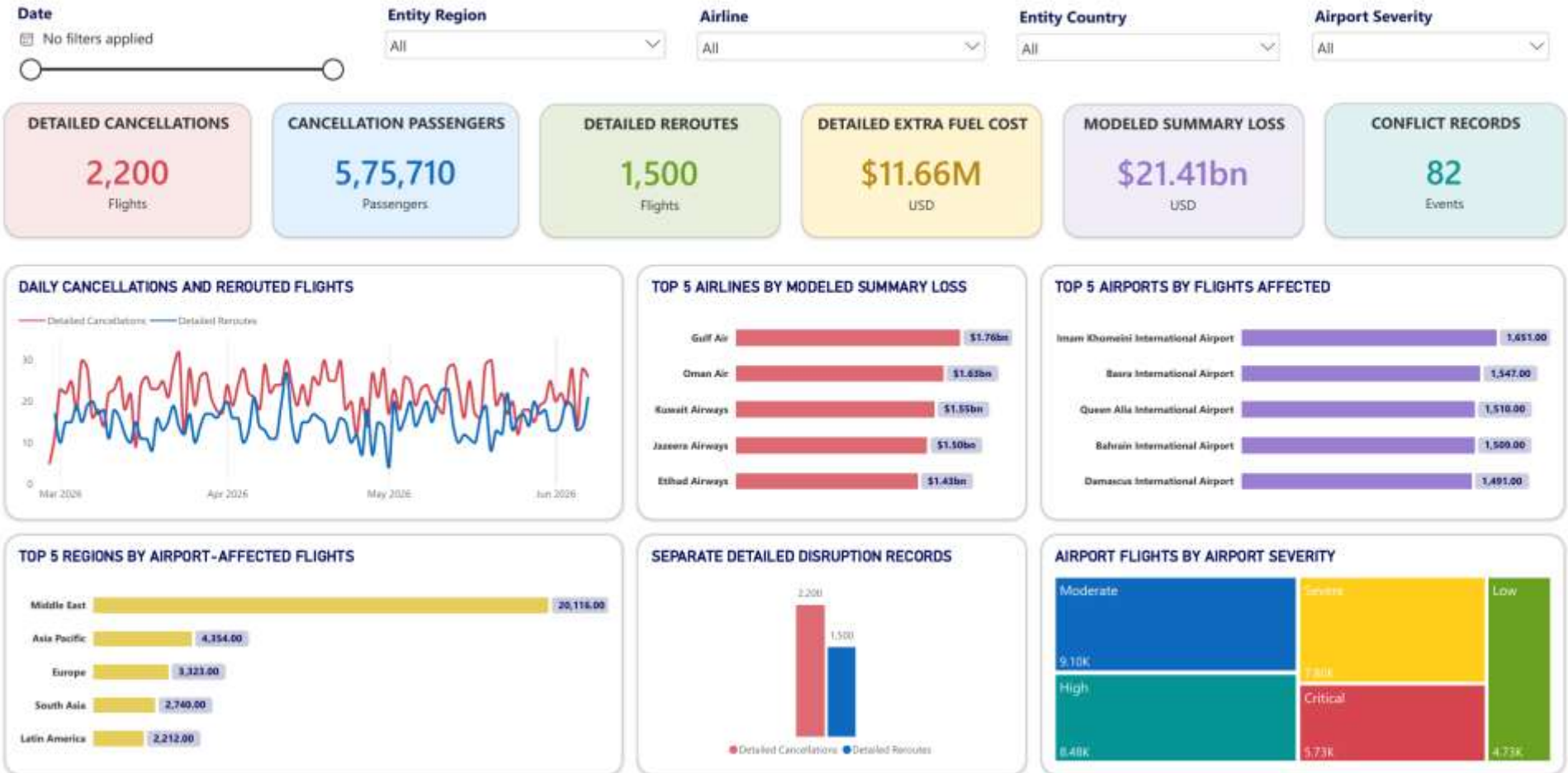
6 Completed Power BI report

The four pages below show the final report with no filters applied. The same measures can be reviewed by the applicable date, airline, country, region, and severity selections. Source-specific filter behavior is documented in the technical reference.

Figure 1 Executive Overview

GLOBAL AVIATION DISRUPTION ASSESSMENT 2026

Executive Overview | Operational, Financial & Geopolitical Impact



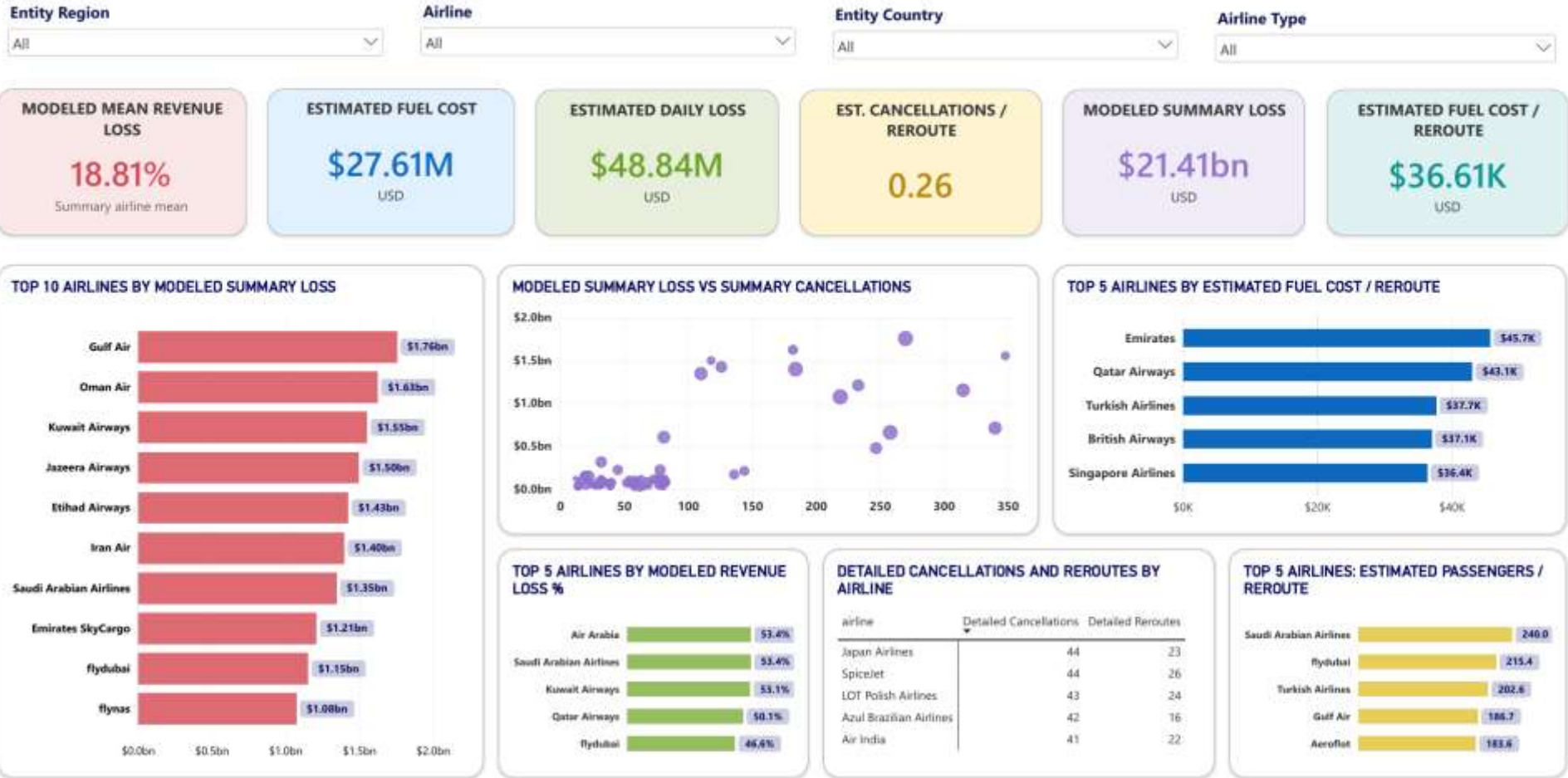
Operational records cover 27 February–7 June 2026 | Conflict events cover a wider period | Static simulated data | Detailed, summary and estimate populations differ | Financial values are undated | Association is not causation.

Shows the operational totals, airport concentration, and financial overview. The cancellation and reroute records are displayed separately.

Figure 2 Modeled Financial Scenarios

GLOBAL AVIATION DISRUPTION ASSESSMENT 2026

Modeled Financial Scenarios



MODELED MEAN REVENUE LOSS

18.81%

Summary airline mean

ESTIMATED FUEL COST

\$27.61M

USD

ESTIMATED DAILY LOSS

\$48.84M

USD

EST. CANCELLATIONS / REROUTE

0.26

MODELED SUMMARY LOSS

\$21.41bn

USD

ESTIMATED FUEL COST / REROUTE

\$36.61K

USD

TOP 10 AIRLINES BY MODELED SUMMARY LOSS

Gulf Air	\$1.76bn
Oman Air	\$1.63bn
Kuwait Airways	\$1.55bn
Jazeera Airways	\$1.50bn
Etihad Airways	\$1.43bn
Iran Air	\$1.40bn
Saudi Arabian Airlines	\$1.35bn
Emirates SkyCargo	\$1.21bn
flydubai	\$1.15bn
flynas	\$1.08bn

MODELED SUMMARY LOSS VS SUMMARY CANCELLATIONS

TOP 5 AIRLINES BY ESTIMATED FUEL COST / REROUTE

Emirates	\$45.7K
Qatar Airways	\$43.1K
Turkish Airlines	\$37.7K
British Airways	\$37.1K
Singapore Airlines	\$36.4K

TOP 5 AIRLINES BY MODELED REVENUE LOSS %

Air Arabia	53.4%
Saudi Arabian Airlines	53.4%
Kuwait Airways	53.1%
Qatar Airways	50.1%
flydubai	46.6%

DETAILED CANCELLATIONS AND REROUTES BY AIRLINE

airline	Detailed Cancellations	Detailed Reroutes
Japan Airlines	44	23
SpiceJet	44	26
LOT Polish Airlines	43	24
Azul Brazilian Airlines	42	16
Air India	41	22

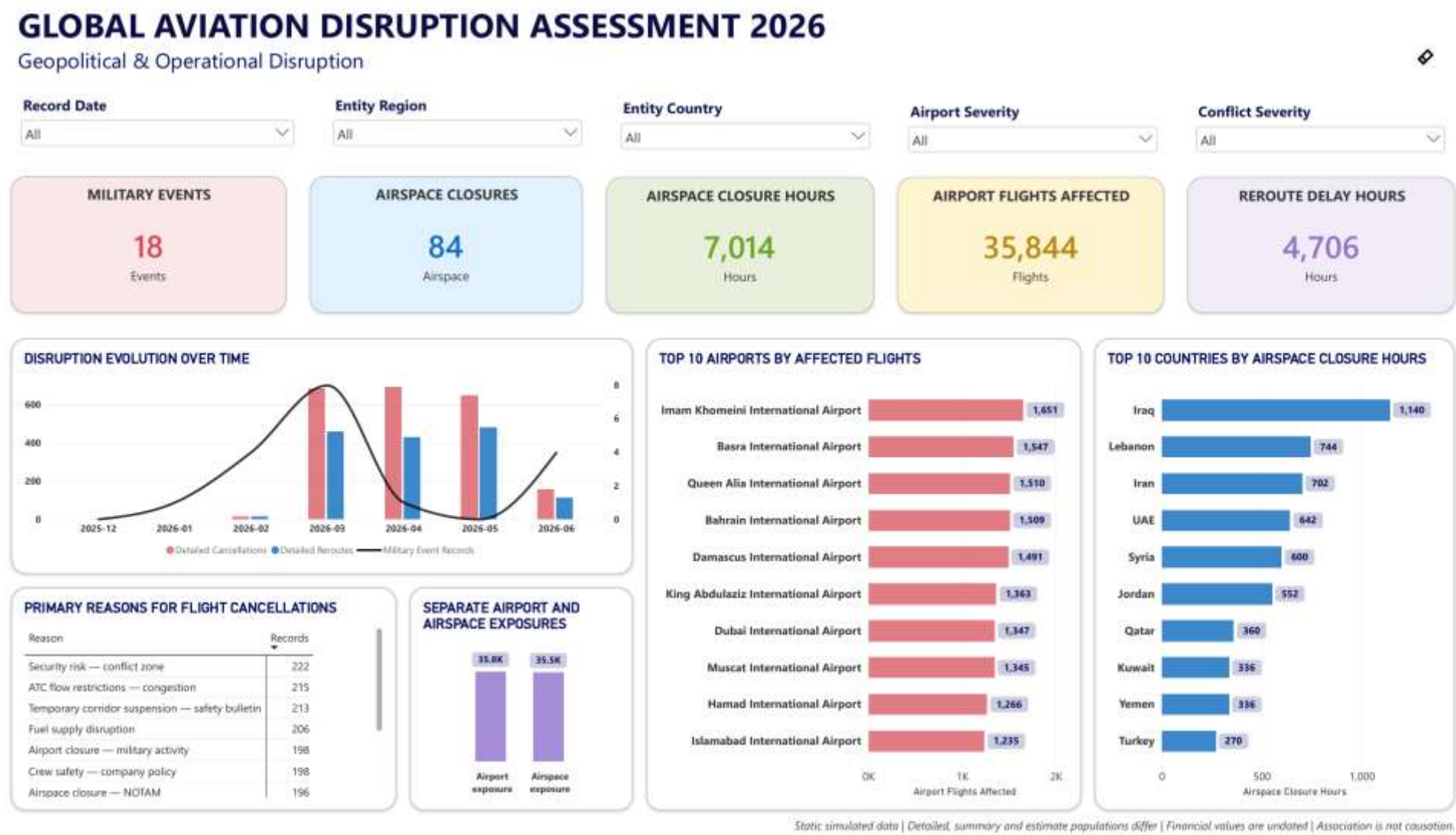
TOP 5 AIRLINES: ESTIMATED PASSENGERS / REROUTE

Saudi Arabian Airlines	240.0
flydubai	215.4
Turkish Airlines	202.6
Gulf Air	186.7
Aeroflot	183.6

Static simulated data | Detailed, summary and estimate populations differ | Financial values are undated | Association is not causation

Separates modeled summary loss from the daily-estimate population and its ratios. Financial values are undated.

Figure 3 Geopolitical and Operational Disruption



Shows operational timelines, supplied cancellation reasons, airport exposure, and separate closure hours.

Figure 4 Regional Risk and Aviation Corridor Vulnerability

GLOBAL AVIATION DISRUPTION ASSESSMENT 2026

Regional Risk & Aviation Corridor Vulnerability

Regional exposure | Country impact | Corridor rerouting | Operational burden | Financial consequence



Static simulated data | Detailed, summary and estimate populations differ | Financial values are undated | Association is not causation

Shows directional corridor burdens and separate airport and airspace exposures. Corridor direction changes the ranking.

7 Recommendations

Focus regional contingency reviews on the Middle East

The region holds 56.12% of airport exposure and 77.85% of modeled summary loss. Start the scenario review there, then assess other regions using their own operational and financial measures.

Prepare corridor plans by direction

Europe to Middle East leads reroute activity and burden; the reverse direction has the most cancellation records. Review alternative routing and passenger-support needs separately for each direction. The analysis identifies planning priorities, not a proven optimal route.

Review high exposure airports and long disruptions separately

Use Imam Khomeini, Basra, and Queen Alia for affected-flight concentration review. Also review the longest individual disruption records, since duration and total flight exposure rank different airports.

Plan passenger assistance around cancellation concentrations

Use the 575,710 cancellation passengers and corridor passenger totals to structure rebooking, communication, and assistance scenarios. Do not allocate reroute passenger capacity using cancellation passengers.

Keep financial scenarios separate in budget discussions

Use summary loss for the 70-airline exposure comparison and daily estimates for the separate 35-airline scenario. Do not convert daily estimates into a period total without a supported period and assumptions.

Review airspace contingency alongside airport planning

Use Iraq, Lebanon, and Iran closure-duration rankings to prioritize scenario review. Keep airport and airspace exposure separate so overlapping flights do not become an inflated total.

Use event context without claiming a causal result

The shared-window comparison does not show a consistent increase across operational measures. Treat event timing as context and use direct operational burdens to support planning priorities.

8 Conclusion

The assessment identifies clear concentrations in Middle East airport and financial exposure, Europe-Middle East corridors, and a small group of high-exposure airports and closure areas. The completed report provides a consistent basis for discussing contingency priorities while keeping detailed operations and modeled financial scenarios separate. The recommended actions are planning proposals based on those findings, not measured improvements already achieved.

Appendix Business question results

The following answers connect the requested analysis to the completed SQL and Power BI outputs. Numbers use the full source population unless a period is stated. The Technical Documentation contains exact formulas and source definitions.

Q01 How many detailed cancellation records are reported

2,200 detailed cancellation records.

Result reference: SQL 06; Executive Overview.

Q02 How many passengers are affected by those cancellations

575,710 passengers in cancellation records.

Result reference: SQL 06; Executive Overview.

Q03 How do cancellations vary over the available dates

Monthly counts: February 17, March 684, April 692, May 650, June 157. February and June have partial coverage.

Result reference: Executive Overview daily trend; cancellation fact.

Q04 Which cancellation reasons are most frequent

Security risk related to a conflict zone: 222; air traffic control congestion: 215; temporary corridor suspension: 213.

Result reference: Geopolitical page cancellation reasons.

Q05 How many detailed reroute records are reported

1,500 detailed reroute records.

Result reference: SQL 06; Executive and Regional pages.

Q06 What additional distance do those reroutes create

2,201,390 additional kilometres.

Result reference: SQL 06; Regional page.

Q07 What additional fuel cost do those reroutes create

USD 11,658,572.59 additional fuel cost.

Result reference: SQL 06; Executive and Regional pages.

Q08 What delay burden do those reroutes create

4,706.30 delay hours.

Result reference: SQL 06; Geopolitical and Regional pages.

Q09 Which directional corridors have the most reroutes

Europe to Middle East: 196; Middle East to Europe: 168.

Result reference: SQL 05 route facts; Regional corridor ranking.

Q10 Which corridors have the most cancellations and affected passengers

Middle East to Europe has 251 cancellations; Europe to Middle East has 250 and the largest regional-corridor passenger sum, 67,417.

Result reference: SQL 05 route aggregation.

Q11 Which corridors carry the greatest additional distance, fuel cost, and delay

Europe to Middle East leads regional-corridor additional distance, fuel cost, and delay. Exact values appear in section 3.2.

Result reference: SQL 05 route measures; Regional page.

Q12 Which regions have the highest airport-affected flight counts

Middle East: 20,116 airport-affected flights, 56.12% of the airport total.

Result reference: SQL 01; Executive region ranking.

Q13 Which countries have the highest airport-affected flight counts

UAE: 2,539 airport-affected flights, followed by Saudi Arabia with 2,521.

Result reference: SQL 01 country ranking; Regional page.

Q14 Which airports have the highest reported disruption exposure

Imam Khomeini International Airport: 1,651; Basra: 1,547; Queen Alia: 1,510.

Result reference: SQL 01; Executive and Geopolitical pages.

Q15 Which airports have the longest reported disruption records

São Paulo-Guarulhos has the longest individual reported disruption, 71.7 hours. This is not the highest aggregate flight exposure.

Result reference: SQL 01 maximum duration by airport.

Q16 How does airport disruption vary over time

March has 12,085 airport-affected flights; April 9,792; May 11,298. February and June cover partial months.

Result reference: SQL 01 daily airport output aggregated by month.

Q17 Which countries have the greatest airspace closure duration

Iraq: 1,140 closure hours; Lebanon: 744; Iran: 702.

Result reference: Airspace detail; Geopolitical country ranking.

Q18 How do airport and airspace exposure differ by region when reported separately

The Middle East has the largest exposure in both sources: 20,116 airport-affected flights and 4,523.1 airport disruption hours, compared with 33,509 airspace-affected flights and 6,384 closure hours. Europe reports 3,323 airport-affected flights and 796.7 hours, compared with 607 airspace-affected flights and 240 closure hours. These contrasting regional profiles support separate airport and airspace contingency reviews. The sources may overlap, so their flight counts and durations must not be added into one deduplicated total.

Result reference: SQL 05 regional view; Regional page.

Q19 Which airlines have the highest modeled total loss

Gulf Air: USD 1.76bn modeled loss; Oman Air: USD 1.63bn; Kuwait Airways: USD 1.55bn.

Result reference: SQL 02; Financial page.

Q20 Which regions have the highest modeled total loss

Middle East: USD 16.67bn, 77.85% of modeled summary loss.

Result reference: SQL 02; Regional page.

Q21 How does modeled loss differ across airline types

Flag Carrier: USD 13.81bn; Low Cost: USD 5.19bn; Cargo: USD 1.71bn; Private: USD 0.71bn.

Result reference: SQL 02 airline-type aggregation.

Q22 Which airlines have the highest estimated daily loss and related exposure

Emirates leads estimated daily loss at USD 4.20M. Its estimate record also reports 18 cancellations, 62 reroutes, USD 2,835,200 in additional fuel cost and 9,180 impacted passengers. Qatar Airways follows at USD 3.10M, with 12 cancellations, 41 reroutes, USD 1,765,800 in fuel cost and 6,540 passengers. El Al is third at USD 2.80M, with 12 cancellations, no reroutes, no additional fuel cost and 3,980 passengers. These are values from the same undated estimate population, not detailed flight totals; reroute-based ratios remain blank where reroutes are zero.

Result reference: SQL 03 daily estimate ranking.

Q23 How do the summary and estimate airline populations differ

70 summary airlines; 35 estimate airlines; 32 shared; 38 summary-only and three estimate-only.

Result reference: SQL 03 full outer join coverage.

Q24 How do daily operational counts differ between military-event and other dates within shared coverage

Across 100 shared dates, mean cancellations are 21.08 on 12 military-event dates and 22.07 on 88 other dates; mean reroutes are 15.08 and 14.99. Mean airport records per day are 1.83 and 2.31.

Result reference: SQL 04 shared operational window.